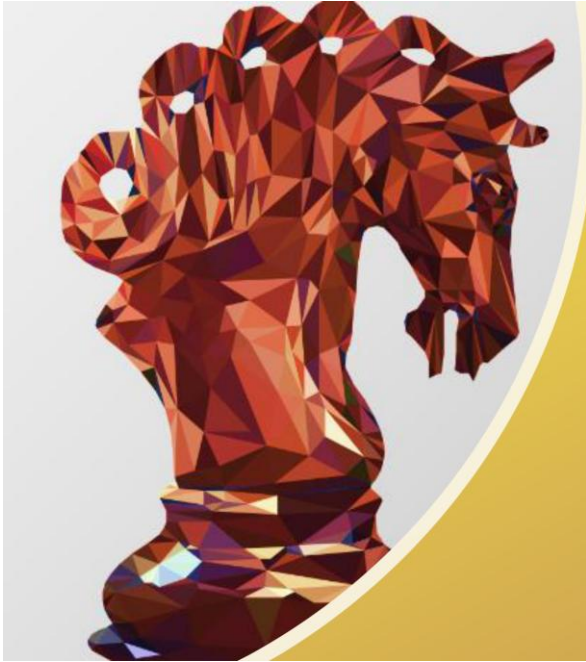




**PRELIMINARY NATIONAL-LEVEL ASSESSMENT**

# **RISKS, VULNERABILITIES, AND THREAT VECTORS AFFECTING THE U.S. ELECTORAL PROCESS**

*Voting Integrity, System Resilience,  
Demonstrated Exploitation, and U.S.-Controlled Risk  
Reduction*

**PUBLIC-SOURCE DRAFT | JULY 2026**

## **DRAFT STATUS**

This document was generated with support of the Tornberg Analyst GPT, The Tornberg Consulting Methodology, and the Tornberg Consulting Analytic Framework. It has not been approved or released by Tornberg Consulting, LLC. The purpose of this draft is analytic review, coordination, and administrative review against analytic, tradecraft, and publication standards.

# Publication and Review Status

This product is a preliminary public-source assessment prepared for internal analytic review.

## DRAFT STATUS

This document was generated through use of the Tornberg Analyst GPT. It has not been reviewed, approved, or released by Tornberg Consulting, LLC. It does not represent work performed directly by Tornberg Consulting, LLC. An accountable human professional must validate the evidence, judgments, recommendations, legal characterizations, and release conditions before use.

## Purpose

Assess risks, vulnerabilities, and threat vectors affecting the U.S. electoral process as a whole and voting integrity in particular; quantify each vector through meaningful, demonstrable, and—where possible—actionable measures; identify unresolved gaps; and evaluate practical risk-reduction measures within lawful U.S. control.

## Scope and time horizon

Primary operational horizon: the 2026 federal election cycle through final certification. Structural vulnerabilities and reforms are considered through 2028 where implementation requires more than one cycle. Public, declassified, judicial, technical, administrative, and high-quality investigative sources are used. No classified access is represented.

## Use limitations

- **Preliminary status.** Unresolved gaps are displayed rather than treated as release defects.
- **No national prevalence claim.** Case repositories and prosecutions establish demonstrated pathways but do not form a representative national incident census.
- **Legal variation.** Eligibility, registration, ballot handling, audits, offenses, and remedies vary by jurisdiction. Recommendations require jurisdiction-specific legal review.
- **Source packet boundary.** This assessment contains only the source qualifications necessary to understand confidence. Detailed provenance, coding, deconfliction, falsification, and method records belong in the companion process summary and source packet.

# Executive Assessment

Preliminary national judgment on election security and voting integrity.

## BOTTOM LINE

The United States faces multiple distinct and demonstrably exploitable election-security threat vectors. The public record reviewed does not establish nationwide compromise of certified vote counts. It does establish successful or attempted exploitation of registration, eligibility, identity, mail-ballot, trusted-access, physical-custody, cyber, reporting, financial, and information pathways. The principal systemic exposure is uneven assurance across a decentralized environment: controls exist, but their coverage, performance, auditability, and detectability are not measured or enforced through one common national framework.

The most practical national risk-reduction opportunity is to establish strong preventive assurance before any registration, ballot, user, software component, or election transaction enters an active process, then reinforce prevention with detection, containment, independent verification, and recovery. For voter registration, the control objective should be to verify under applicable law that each applicant satisfies every legally material eligibility criterion before activating a record capable of ordinary ballot issuance, while preserving due process, alternative evidence, secondary review, cure, appeal, and provisional participation when authoritative information is incomplete.

Independent reconstruction remains indispensable. No critical election function should rely on a technology platform both to perform the function and to provide the only evidence that it was performed correctly. Voter-verifiable paper records, ballot accounting, controlled custody, meaningful post-election audits, standardized event logging, software validation, manual contingencies, and authenticated public communication create independent evidence when preventive controls fail.

Confidence: Moderate to high. Confidence is high that the identified vectors and uneven control environment exist. Confidence is lower concerning undetected prevalence, recurrence, actor coordination, scalability, and aggregate electoral effect because incident reporting, legal definitions, control-performance measures, audit practices, and detection probabilities remain inconsistent.

## Priority judgments

| # | PRIORITY JUDGMENT                                                                                                                                                                                                                                                                                                                                                                           | CONFIDENCE    |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1 | Multiple election-security vectors are real and warning-relevant because documented cases, official findings, direct observation, and technical reporting show that several pathways have been exploited or credibly attempted. <b>A validated instance establishes operational exploitability; it does not establish national prevalence, recurrence, coordination, or outcome effect.</b> | HIGH          |
| 2 | Pre-entry assurance is under-standardized. Registration and eligibility controls should test each legally material criterion before ordinary activation, while measuring and reducing both improper activation and improper exclusion.                                                                                                                                                      | MODERATE-HIGH |
| 3 | Paper records are now nearly universal, but paper availability is not equivalent to independent outcome assurance. Audit design, ballot accounting, custody, escalation, and certification practices remain uneven, and national audit standards remain voluntary or jurisdiction-dependent.                                                                                                | HIGH          |
| 4 | Technology improves capacity and accessibility but creates concentration, credential, vendor, logging, software, and common-mode failure risk when the platform performing a function is also the sole source of proof.                                                                                                                                                                     | HIGH          |
| 5 | Foreign actors can exploit domestic data, process weaknesses, authentic disputes, and public uncertainty without altering a recorded vote. The PRC baseline supports sustained voter-data collection and bounded influence capability; it does not establish PRC alteration of ballots, tabulation, certification, or the 2020 result.                                                      | MODERATE-HIGH |
| 6 | The available evidence cannot reliably rank vectors by national frequency or undetected prevalence. The principal analytic gap has shifted from whether vectors exist to how often they are attempted, when controls fail, what remains invisible, and how far exploitation can progress.                                                                                                   | HIGH          |

Principal sources: EAC 2024 EAVS [EAC-01]; EAC audit and technology guidance [EAC-02–06]; NIST Election Infrastructure Profile [NIST-01]; National Academies [NAS-01]; CISA/ODNI foreign influence guidance [FMI-01–02]; DOJ records [DOJ-01–03]; prior controlled PRC baseline [TB-PRC-01].

# 1. Analytical Scope and Quantification

## 1.1 Governing analytical rule

|                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DO NOT COMPRESS STAGES</b><br>Vulnerability → Attempt → Access or transaction → Acceptance → Count or operational effect → Certification effect → Outcome effect. Evidence that a pathway reached one stage is not proof that it reached every later stage. Conversely, incomplete public evidence at a later stage does not establish that the stage did not occur. |                                                                                                                                                                                                                                                                           |
| <b>BASELINE FINDING</b><br>A proposition supported by evidence sufficiently mature for the bounded claim stated. Example: duplicate registration transactions occur at substantial administrative scale.                                                                                                                                                                | <b>WARNING JUDGMENT</b><br>A reasoned judgment that a demonstrated or credible pathway deserves attention before prevalence or effect is fully known. Example: active duplicate records combined with weak cross-jurisdiction controls can enable multiple-ballot access. |

**ANALYST NOTE:** One validated exploitation event is sufficient to establish that a vulnerability is operationally exploitable and may justify warning, mitigation, or an indicator. It is not sufficient by itself to establish widespread activity, a national rate, likely recurrence, adversary coordination, counted-ballot effect, or outcome determination.

## 1.2 Quantification standard

The assessment does not use a single composite risk score. A numerical score would create false comparability among unlike vectors and could imply precision unsupported by the evidence. Each vector is quantified through the most decision-useful observable measures available.

| DIMENSION             | DECISION USE                                                                                                            |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Exposure              | Size of the susceptible population, transaction set, system, jurisdiction, workforce, or privilege group.               |
| Observed exploitation | Verified cases, convictions, official findings, directly observed transactions, intrusions, or control activations.     |
| Control coverage      | Proportion of exposure protected by a defined preventive, detective, or assurance control.                              |
| Control performance   | False-acceptance, false-rejection, detection, cure, reconciliation, correction, recovery, or audit-escalation rate.     |
| Detectability         | Likelihood that attempted or successful exploitation would create, preserve, and receive review of observable evidence. |
| Consequence stage     | Farthest supported stage reached by the evidence ladder.                                                                |
| Warning indicator     | Observable condition that should trigger attention, collection, preparation, or action.                                 |
| Gap burden            | Material G-01 through G-10 gaps limiting a stronger judgment.                                                           |
| Confidence            | Strength of evidence for the specific proposition—not a substitute for likelihood or consequence.                       |

## 1.3 Frequency language

“Most common” means most frequent *within the identified dataset, period, unit, inclusion rules, and coding scheme* unless a representative national basis is independently established. A case repository, prosecution series, or advocacy database may establish demonstrated methods and generate collection leads; it does *not* establish national prevalence without a defined denominator, known detection probability, consistent dispositions, deduplication, and representative coverage.

1.4 Gap codes

|                                          |                                                            |                                           |                                                      |                                                   |
|------------------------------------------|------------------------------------------------------------|-------------------------------------------|------------------------------------------------------|---------------------------------------------------|
| <b>G-01</b><br>Completeness and coverage | <b>G-02</b><br>Detection and observability                 | <b>G-03</b><br>Evidence-stage bridge      | <b>G-04</b><br>Control performance and residual risk | <b>G-05</b><br>Legal and procedural normalization |
| <b>G-06</b><br>Exposure and denominator  | <b>G-07</b><br>Actor attribution, intent, and coordination | <b>G-08</b><br>System and vector coverage | <b>G-09</b><br>Warning design                        | <b>G-10</b><br>Source integrity and independence  |

2. National Exposure and Assurance Baseline

2.1 Scale of the administered system

|                           |                        |                                                                                                               |
|---------------------------|------------------------|---------------------------------------------------------------------------------------------------------------|
| REGISTERED AND ELIGIBLE   | 234.5 million          | Reported for the 2024 general election; totals can include records pending lawful removal.                    |
| ACTIVE REGISTRATIONS      | 211.1 million          | About 90 percent of reported registered-and-eligible records.                                                 |
| REGISTRATION TRANSACTIONS | 103+ million           | Applications, updates, duplicates, removals, and related transactions between 2022 and 2024.                  |
| DUPLICATE TRANSACTIONS    | 10.4 million / 12.7%   | Exact-match submissions to existing records; commonly repeated submissions, not evidence of duplicate voting. |
| RECORDS REMOVED           | 21+ million            | List-maintenance removals during the two-year period.                                                         |
| BALLOTS COUNTED           | 158+ million           | 2024 general election; turnout approximately 64.7 percent of citizen voting-age population.                   |
| MAIL VOTING               | ~30% of ballots        | About 15 million mail ballots were returned through drop boxes in 36 states.                                  |
| PAPER EVIDENCE            | >98% of jurisdictions  | Paper ballot or auditable paper record available; paper alone does not establish meaningful audit.            |
| ELECTRONIC POLLBOOKS      | 39.9% of jurisdictions | Used in 41 states; testing and certification requirements remain uneven.                                      |
| ELECTION-DAY WORKFORCE    | 675,434 workers        | 94,466 were new among the population for which tenure was reported.                                           |

Source: 2024 Election Administration and Voting Survey [EAC-01]. EAVS is the strongest national administrative dataset in this assessment. It relies on certified state and local submissions, but completeness, definitions, and reporting capability vary by item and jurisdiction.

2.2 What these measures establish

- **Scale.** Registration, ballot, personnel, technology, and custody processes operate at national scale and create large exposure populations.
- **Routine complexity.** Duplicate registration transactions, list maintenance, ballot curing, and record corrections are normal administrative activity. An anomaly count is not a fraud count.
- **Control activation.** Rejected, cured, corrected, or reconciled transactions often show safeguards working; they should not be classified automatically as successful exploitation.
- **Uneven assurance.** Paper evidence is broadly available, but audit design, pollbook certification, incident logging, and technology standards differ substantially.
- **Measurement limits.** National exposure can often be quantified; national exploitation, detection, and residual risk usually cannot.

## 2.3 Principal controllable vulnerabilities

|                                                                                                                                                                                                                                                            |                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRE-ENTRY UNCERTAINTY</b><br><p>No common national outcome standard requires every legally material registration criterion to be affirmatively supported before ordinary activation, with uniform alternative evidence and due-process protections.</p> | <b>CONTROL OBJECTIVE</b><br><p>Accurate, timely, reviewable eligibility determination that measures both improper activation and improper exclusion.</p> |
| <b>INDEPENDENT VERIFICATION GAP</b><br><p>Paper records are widespread, but not every material outcome is verified through a statistically meaningful audit independent of the initial tallying system.</p>                                                | <b>CONTROL OBJECTIVE</b><br><p>Reconstruct material outcomes from evidence independent of the system producing the initial result.</p>                   |
| <b>TECHNOLOGY DEPENDENCE</b><br><p>Some functions rely on platforms, vendors, credentials, or software whose logging, testing, remote access, change control, and fallback procedures are uneven.</p>                                                      | <b>CONTROL OBJECTIVE</b><br><p>No critical platform should be both executor and exclusive source of proof.</p>                                           |
| <b>FRAGMENTED DETECTABILITY</b><br><p>Incident taxonomies, referrals, evidence retention, disposition tracking, and control-performance measures are not consistently comparable nationwide.</p>                                                           | <b>CONTROL OBJECTIVE</b><br><p>Standardize what is logged, reported, retained, investigated, and measured.</p>                                           |
| <b>DATA EXPOSURE</b><br><p>Voter and political data support administration but can also enable targeting, identity misuse, coercion, surveillance, and influence.</p>                                                                                      | <b>CONTROL OBJECTIVE</b><br><p>Minimize collection and disclosure, control access, monitor use, and protect sensitive fields.</p>                        |
| <b>COMMUNICATION LATENCY</b><br><p>Delays, unexplained anomalies, and inconsistent public authentication create space for adversaries and domestic actors to exploit genuine uncertainty.</p>                                                              | <b>CONTROL OBJECTIVE</b><br><p>Authenticate official channels, pre-plan incident communications, and publish verified control outcomes.</p>              |

## 2.4 Integrated control model

|                                                                                             |                                                                                    |                                                                                                  |                                                                                                               |                                                                                             |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>1</b><br><b>PREVENT</b><br><p>Stop invalid or unauthorized inputs before activation.</p> | <b>2</b><br><b>DETECT</b><br><p>Identify attempts, errors, and control bypass.</p> | <b>3</b><br><b>CONTAIN / CORRECT</b><br><p>Limit spread, resolve records, preserve evidence.</p> | <b>4</b><br><b>INDEPENDENTLY VERIFY</b><br><p>Reconstruct process and outcome outside originating system.</p> | <b>5</b><br><b>RECOVER / LEARN</b><br><p>Restore operations and remediate the weakness.</p> |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

Figure 1. Integrated election-assurance control model. No layer substitutes for another.

### 3. Threat Vector–Gap Register

The register separates confidence that a vector exists from confidence in its national scale, recurrence, and effect. Metrics are bounded to the source or exposure population stated. “Demonstrated” means at least one supported exploitation or control-bypass event; it does not mean widespread or outcome-determinative activity.

| ID    | THREAT VECTOR                                                         | SUPPORTED EVIDENCE STAGE                                                                      | QUANTITATIVE / ACTIONABLE ANCHOR                                                                                                 | PRIMARY CONTROL EMPHASIS                                                                                                           | GAPS                      | CONFIDENCE                                   |
|-------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------|
| TV-01 | <b>Voter registration and list integrity</b>                          | Vulnerability and documented false-registration exploitation                                  | 234.5m eligible records; 103m+ transactions; 10.4m duplicate transactions (12.7%).                                               | Eligibility validation; application provenance; early deduplication; notice/cure; record audit trail.                              | G-01,02,03,04,05,06,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-02 | <b>Eligibility exploitation</b>                                       | Documented ineligible registrations and voting cases; national rate unresolved                | Track verified cases and dispositions; % applications requiring secondary eligibility review; false acceptance/rejection.        | Criterion-by-criterion validation; lawful evidence alternatives; secondary review; provisional path.                               | G-01,02,03,05,06,07,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-03 | <b>Identity misuse and impersonation</b>                              | Documented identity misuse, forged records, and impersonation                                 | 15 records coded in-person impersonation in 1,619-row Heritage export—dataset only, not national prevalence.                     | Identity resolution; protected data; signature/credential controls; conflict resolution; evidence retention.                       | G-01,02,03,04,06,07,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-04 | <b>Duplicate or double voting</b>                                     | Convictions establish completed multiple-ballot conduct in bounded cases                      | 143 records coded duplicate voting in Heritage export; reconcile matched records, ballot issuance, and voting history.           | Interstate matching; duplicate-ballot suppression; pre-certification reconciliation; cross-jurisdiction referral.                  | G-01,02,03,04,05,06,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-05 | <b>Absentee and mail-ballot exploitation</b>                          | Documented forged, unauthorized, coerced, or improperly returned ballots                      | ~30% of 158m+ ballots cast by mail; 585,457 entered cure; 317,191 cured; 270,753 rejected.                                       | Identity/signature validation; custody and return controls; cure; ballot tracking; issuance/acceptance reconciliation.             | G-01,02,03,04,05,06,09    | <b>Existence HIGH / Scale–effect LOW–MOD</b> |
| TV-06 | <b>Vote buying, coercion, intimidation, and paid registration</b>     | Directly observed and prosecuted paid-registration pathway; other cases adjudicated           | Armstrong federal case; 28 filmed petition/registration exchanges reported by O’Keefe; 156 buying-vote rows in Heritage dataset. | Worker authorization; payment audits; vulnerable-population outreach; rapid referral; evidence preservation.                       | G-01,02,03,04,07,09,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-07 | <b>Petition, nomination, and ballot-access fraud</b>                  | Documented forged signatures, false identities, and improper incentives                       | 103 ballot-petition-fraud rows in Heritage export; case-level disposition and affected-signature counts.                         | Circulator registration; signature validation; controlled voter-data access; random verification; chain of custody.                | G-01,02,03,04,07,08,10    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-08 | <b>Insider and trusted-access misconduct</b>                          | Documented misuse by officials, workers, contractors, and other authorized actors             | 675,434 Election Day workers; 94,466 new among reported workforce; count privileged users and dual-control coverage.             | Least privilege; unique credentials/MFA; two-person control; background/training; logs; rotation and supervision.                  | G-01,02,03,04,07,08,09    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-09 | <b>Cyber compromise of election and campaign systems</b>              | Repeated targeting and access capability; certified vote alteration not established           | 39.9% of jurisdictions use e-pollbooks; measure exposed systems, incidents, MFA/log coverage, dwell and recovery time.           | Segmentation; MFA; patching; allowlisting; monitoring; incident response; manual continuity; independent records.                  | G-01,02,03,04,06,07,08,09 | <b>Attempts HIGH / Count effect LOW</b>      |
| TV-10 | <b>Vendor, equipment, software, and supply-chain compromise</b>       | Technical vulnerabilities and uneven assurance; altered U.S. counts not established           | >98% paper evidence; among states reporting e-pollbook use, 39.5% reported no required pre-purchase test/certification.          | Certification; software provenance/hash validation; SBOM; remote-access limits; vendor logging; acceptance testing.                | G-01,02,03,04,06,07,08,09 | <b>Vulnerability HIGH / Effect LOW</b>       |
| TV-11 | <b>Physical custody, ballot stuffing, destruction, and disruption</b> | Documented physical offenses and operational threats                                          | ~15m mail ballots returned via drop boxes in 36 states; measure custody exceptions and inventory reconciliation.                 | Sealed containers; bipartisan/two-person transfer; tamper evidence; cameras where lawful; inventory and route logs.                | G-01,02,03,04,05,06,09    | <b>Existence HIGH / Scale–effect LOW</b>     |
| TV-12 | <b>Reporting, canvass, recount, and certification disruption</b>      | Demonstrated reporting errors and procedural obstruction; distinct from tabulation compromise | Correction latency; unresolved discrepancies; audit escalations; missed statutory deadlines; certification exceptions.           | Separate unofficial from certified systems; authenticated feeds; reconciliation; escalation rules; continuity and legal playbooks. | G-01,02,03,04,05,07,08,09 | <b>Existence MOD / Scale–effect LOW</b>      |

| ID    | THREAT VECTOR                                         | SUPPORTED EVIDENCE STAGE                                                                                             | QUANTITATIVE / ACTIONABLE ANCHOR                                                                                            | PRIMARY CONTROL EMPHASIS                                                                                                   | GAPS                      | CONFIDENCE                              |
|-------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------|
| TV-13 | Information operations and public-confidence attack   | Repeated foreign and domestic exploitation of authentic disputes, false claims, personas, leaks, and synthetic media | Attributed campaigns; impersonated officials; audience reach; correction latency; administrative or behavioral effects.     | Authenticated .gov channels; prebunking; provenance; rapid incident communications; monitoring; trusted validators.        | G-01,02,03,06,07,09,10    | Existence HIGH / Recurrence MOD-HIGH    |
| TV-14 | Illicit campaign finance and covert political support | Documented offense class; current collected corpus incomplete                                                        | Prohibited contributions, straw-donor cases, beneficial-ownership gaps, referrals, disposition and funds traced.            | Donor identity validation; beneficial ownership; payment analytics; foreign-source screening; referral and disclosure.     | G-01,02,03,05,06,07,08,10 | Existence MOD / Scale-effect LOW        |
| TV-15 | Systemic, accidental, and cascading failure           | Frequent nonmalicious errors and outages; may be amplified or deliberately exploited                                 | Outages, configuration errors, delayed openings, correction time, manual-contingency coverage, common-vendor concentration. | Testing; change control; redundancy; training; manual fallbacks; exercises; after-action review; authenticated correction. | G-01,02,04,06,08,09       | Existence HIGH / Event-specific LOW-MOD |

Table 1. Threat Vector–Gap Register summary. Confidence statements are proposition-specific. Counts from the Heritage export describe that controlled dataset only and are not national prevalence estimates.

### 3.1 Vector records

#### TV-01 Voter registration and list integrity

**EVIDENCE-BACKED BASELINE:** Duplicate, stale, inaccurate, and fraudulent records occur. EAVS reports 10.4 million exact-match duplicate transactions, most commonly repeated submissions. Directly observed and prosecuted false-registration conduct establishes exploitability.

**WARNING JUDGMENT:** Weak identity resolution, false addresses, stale records, or poor provenance can create later ballot-access, identity, targeting, or public-confidence exposure.

**DEMONSTRABLE / ACTIONABLE METRIC:** 234.5m eligible records; 103m+ transactions; 10.4m duplicate transactions (12.7%).

**PREVENTIVE CONTROL:** Validate each legal criterion before ordinary activation; preserve application provenance; run intrastate and available interstate deduplication; use fair alternative evidence and secondary review.

**FORENSIC / ASSURANCE CONTROL:** Audit record creation and changes; reconcile ballot issuance and voting history; preserve disposition and correction records.

**INDICATORS:** Registration-validation coverage; unresolved duplicate rate; median resolution; improper activation/exclusion.

**PRINCIPAL GAPS:** G-01,02,03,04,05,06,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

#### TV-02 Eligibility exploitation

**EVIDENCE-BACKED BASELINE:** Voting by persons ineligible under applicable law is a documented offense category. Eligibility rules and restoration processes vary, and aggregate case repositories cannot establish a national rate.

**WARNING JUDGMENT:** Inconsistent source data or weak review can permit improper activation; overly rigid matching can also exclude eligible citizens and create an integrity failure of a different kind.

**DEMONSTRABLE / ACTIONABLE METRIC:** Track verified cases and dispositions; % applications requiring secondary eligibility review; false acceptance/rejection.

**PREVENTIVE CONTROL:** Criterion-by-criterion validation for identity, citizenship where required, age, residence, jurisdiction, disqualification, restoration, and duplicate status; no single-database veto.

**FORENSIC / ASSURANCE CONTROL:** Measure false acceptance, false rejection, cure, appeal, reversal, and provisional-ballot outcomes; retain the evidence supporting every adverse decision.

**INDICATORS:** Secondary-review rate; cure/appeal/reversal; false acceptance and false rejection; provisional resolution.

**PRINCIPAL GAPS:** G-01,02,03,05,06,07,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

#### TV-03 Identity misuse and impersonation

**EVIDENCE-BACKED BASELINE:** Identity misuse, forged voter or petition records, and in-person impersonation have occurred. The Heritage export contains 15 records coded in-person impersonation; that is a dataset count, not a national rate.

**WARNING JUDGMENT:** Public or illicitly obtained voter data can enable forged records, impersonation, targeted coercion, or false evidence even when no fraudulent ballot is counted.

**DEMONSTRABLE / ACTIONABLE METRIC:** 15 records coded in-person impersonation in 1,619-row Heritage export—dataset only, not national prevalence.

**PREVENTIVE CONTROL:** Minimize exposed data; verify identity using lawful methods; control access to signatures and protected fields; require provenance for worker-entered changes.

**FORENSIC / ASSURANCE CONTROL:** Preserve images, signatures, access records, device logs, and case dispositions; compare challenged identity events with legitimate voter participation.

**INDICATORS:** Protected-field access anomalies; identity conflicts; signature or form forgeries; verified dispositions.

**PRINCIPAL GAPS:** G-01,02,03,04,06,07,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-04 Duplicate or double voting**

**EVIDENCE-BACKED BASELINE:** Completed double voting is established in bounded convictions. Duplicate registration alone is not evidence that multiple ballots were cast, accepted, or counted.

**WARNING JUDGMENT:** Active records in more than one jurisdiction combined with weak issuance and voting-history reconciliation create an operational pathway for multiple ballots.

**DEMONSTRABLE / ACTIONABLE METRIC:** 143 records coded duplicate voting in Heritage export; reconcile matched records, ballot issuance, and voting history.

**PREVENTIVE CONTROL:** Search for active duplicates before ballot issuance; share high-confidence matches lawfully; suppress duplicate issuance only under controlled and reviewable rules.

**FORENSIC / ASSURANCE CONTROL:** Reconcile ballots issued, returned, accepted, rejected, and counted across matched records before certification; record final dispositions and false matches.

**INDICATORS:** High-confidence matched voting histories; duplicate issuance; reconciliation completion; case disposition.

**PRINCIPAL GAPS:** G-01,02,03,04,05,06,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-05 Absentee and mail-ballot exploitation**

**EVIDENCE-BACKED BASELINE:** Mail-ballot fraud is documented through forged, unauthorized, coerced, or improperly returned ballots. In 2024, about 30 percent of ballots were cast by mail; 585,457 entered cure, more than half were cured, and 270,753 were rejected.

**WARNING JUDGMENT:** Large-scale remote issuance and return create identity, custody, assistance, coercion, and acceptance pathways; rejection and cure data show both exposure and control activity.

**DEMONSTRABLE / ACTIONABLE METRIC:** ~30% of 158m+ ballots cast by mail; 585,457 entered cure; 317,191 cured; 270,753 rejected.

**PREVENTIVE CONTROL:** Control ballot issuance; validate request and identity; provide secure tracking; limit unauthorized handling; design clear assistance and collection rules.

**FORENSIC / ASSURANCE CONTROL:** Reconcile issue-return-accept-count; preserve envelope and cure evidence; monitor rejection reasons; audit custody and duplicate-ballot controls.

**INDICATORS:** Request/return/accept/reject/cure rates; custody exceptions; duplicate-ballot control activations.

**PRINCIPAL GAPS:** G-01,02,03,04,05,06,09

**CONFIDENCE:** Existence HIGH / Scale-effect LOW-MOD

**TV-06 Vote buying, coercion, intimidation, and paid registration**

**EVIDENCE-BACKED BASELINE:** Vote buying, coercion, intimidation, and paid registration are documented. The Armstrong case, supported by video and federal action, established payment for registration and false-address exposure; broader recorded petition activity supplies additional direct observation.

**WARNING JUDGMENT:** Economic incentives and coercion can corrupt participation upstream of counting and can target vulnerable populations without leaving ordinary tabulation anomalies.

**DEMONSTRABLE / ACTIONABLE METRIC:** Armstrong federal case; 28 filmed petition/registration exchanges reported by O'Keefe; 156 buying-vote rows in Heritage dataset.

**PREVENTIVE CONTROL:** Authorize and monitor registration/petition workers; prohibit and audit incentives; provide protected reporting; conduct targeted outreach to vulnerable populations.

**FORENSIC / ASSURANCE CONTROL:** Preserve video, payment, worker, form, and referral records; map cases to registrations, ballots, petitions, and final dispositions without assuming downstream effect.

**INDICATORS:** Worker or address concentration; payment evidence; complaints; referrals and final legal disposition.

**PRINCIPAL GAPS:** G-01,02,03,04,07,09,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-07 Petition, nomination, and ballot-access fraud**

**EVIDENCE-BACKED BASELINE:** Forged petition signatures, false identities, and improper incentives are documented. The Heritage export contains 103 ballot-petition-fraud rows; O'Keefe reported 28 recorded exchanges across petition and registration activity.

**WARNING JUDGMENT:** Compromised voter data and paid circulator incentives can manipulate ballot access and candidate or initiative qualification independently of voter fraud.

**DEMONSTRABLE / ACTIONABLE METRIC:** 103 ballot-petition-fraud rows in Heritage export; case-level disposition and affected-signature counts.

**PREVENTIVE CONTROL:** Register and train circulators where lawful; control voter-data access; establish payment safeguards; validate random or risk-selected signatures early.

**FORENSIC / ASSURANCE CONTROL:** Preserve petition custody, validation decisions, source devices, and worker identifiers; conduct reproducible sampling and case-level adjudication.

**INDICATORS:** Circulator outliers; sample validation failures; identity reuse; unauthorized voter-data access.

**PRINCIPAL GAPS:** G-01,02,03,04,07,08,10

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-08 Insider and trusted-access misconduct**

**EVIDENCE-BACKED BASELINE:** Officials, workers, contractors, vendors, and campaign personnel with authorized access have committed election-related offenses. The workforce and privilege population are large and change by cycle.

**WARNING JUDGMENT:** Trusted access can bypass external defenses, suppress evidence, or create high-consequence common-mode failure with fewer observable steps.

**DEMONSTRABLE / ACTIONABLE METRIC:** 675,434 Election Day workers; 94,466 new among reported workforce; count privileged users and dual-control coverage.

**PREVENTIVE CONTROL:** Least privilege; unique credentials and MFA; background and role screening; two-person control; separation of duties; supervised temporary access.

**FORENSIC / ASSURANCE CONTROL:** Immutable or independently retained logs; access review; reconciliation by personnel outside the originating function; protected insider reporting.

**INDICATORS:** Privileged-user inventory; MFA and dual-control coverage; access exceptions; log-review completion.

**PRINCIPAL GAPS:** G-01,02,03,04,07,08,09

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-09 Cyber compromise of election and campaign systems**

**EVIDENCE-BACKED BASELINE:** Foreign and criminal actors possess and use capabilities relevant to voter registration, pollbooks, reporting, campaigns, and election-office networks. The record **does not** establish foreign alteration of certified U.S. vote totals.

**WARNING JUDGMENT:** Access can support disruption, data theft, influence, false evidence, or delayed administration even when voting equipment is not altered.

**DEMONSTRABLE / ACTIONABLE METRIC:** 39.9% of jurisdictions use e-pollbooks; measure exposed systems, incidents, MFA/log coverage, dwell and recovery time.

**PREVENTIVE CONTROL:** Segment networks; isolate critical systems; require MFA, allowlisting, patching, tested backups, restricted remote access, and manual continuity.

**FORENSIC / ASSURANCE CONTROL:** Centralize and retain logs; validate configurations and hashes; conduct incident response, independent reconciliation, and post-event technical review.

**INDICATORS:** Internet exposure; MFA/log coverage; incidents; dwell time; recovery time; independent reconciliation.

**PRINCIPAL GAPS:** G-01,02,03,04,06,07,08,09

**CONFIDENCE:** Attempts HIGH / Count effect LOW

**TV-10 Vendor, equipment, software, and supply-chain compromise**

**EVIDENCE-BACKED BASELINE:** Voting systems, e-pollbooks, election-management systems, and service providers create vendor and supply-chain dependencies. Paper evidence is widespread, but certification generations and e-pollbook requirements vary.

**WARNING JUDGMENT:** A common vendor, update, credential, or remote service can create multi-jurisdiction exposure; lack of independent evidence can turn a platform failure into an assurance failure.

**DEMONSTRABLE / ACTIONABLE METRIC:** >98% paper evidence; among states reporting e-pollbook use, 39.5% reported no required pre-purchase test/certification.

**PREVENTIVE CONTROL:** Require certification/acceptance testing, software provenance, signed updates, SBOM where feasible, hash validation, remote-access limits, and vendor incident duties.

**FORENSIC / ASSURANCE CONTROL:** Compare installed software with trusted baselines; independently audit logs and paper records; test failover and recover from known-good media.

**INDICATORS:** Certification generation; SBOM/provenance; remote access; hash variance; multi-jurisdiction common failure.

**PRINCIPAL GAPS:** G-01,02,03,04,06,07,08,09

**CONFIDENCE:** Vulnerability HIGH / Effect LOW

**TV-11 Physical custody, ballot stuffing, destruction, and disruption**

**EVIDENCE-BACKED BASELINE:** Ballot stuffing, theft, destruction, and operational disruption are documented categories. Approximately 15 million mail ballots were returned via drop boxes in 36 states in 2024, creating a large custody and logistics process.

**WARNING JUDGMENT:** Physical access or custody failure can affect ballot availability, integrity, secrecy, or lawful participation and may be difficult to reconstruct without complete records.

**DEMONSTRABLE / ACTIONABLE METRIC:** ~15m mail ballots returned via drop boxes in 36 states; measure custody exceptions and inventory reconciliation.

**PREVENTIVE CONTROL:** Use sealed and serialized containers, two-person or bipartisan transfer, tamper evidence, controlled routes, access restrictions, and inventory controls.

**FORENSIC / ASSURANCE CONTROL:** Reconcile every transfer and ballot inventory; preserve custody records and video where lawful; investigate exceptions before certification.

**INDICATORS:** Seal and inventory exceptions; transfer completeness; route deviation; unresolved custody variance.

**PRINCIPAL GAPS:** G-01,02,03,04,05,06,09

**CONFIDENCE:** Existence HIGH / Scale-effect LOW

**TV-12 Reporting, canvass, recount, and certification disruption**

**EVIDENCE-BACKED BASELINE:** Unofficial reporting, canvass, recount, certification, and legal contest are distinct from tabulation. Errors and obstruction can create operational and confidence effects without altering the underlying ballots.

**WARNING JUDGMENT:** An actor can exploit delay, false reporting, procedural ambiguity, or pressure on officials to disrupt acceptance or certification.

**DEMONSTRABLE / ACTIONABLE METRIC:** Correction latency; unresolved discrepancies; audit escalations; missed statutory deadlines; certification exceptions.

**PREVENTIVE CONTROL:** Separate public reporting from certified systems; authenticate feeds; predefine authority, escalation, deadlines, and continuity; rehearse legal and operational playbooks.

**FORENSIC / ASSURANCE CONTROL:** Reconcile unofficial results to canvass; document corrections; audit changes; preserve certification records, dissent, and statutory actions.

**INDICATORS:** Correction latency; variance; audit escalation; certification delay; threats against officials.

**PRINCIPAL GAPS:** G-01,02,03,04,05,07,08,09

**CONFIDENCE:** Existence MOD / Scale-effect LOW

**TV-13 Information operations and public-confidence attack**

**EVIDENCE-BACKED BASELINE:** Foreign and domestic actors have used false claims, authentic incidents taken out of scale, deceptive personas, leaks, synthetic content, and impersonation to target elections and confidence.

**WARNING JUDGMENT:** A foreign actor may achieve strategic effect by exploiting genuine vulnerabilities, stolen data, domestic voices, or reporting delays without altering a ballot.

**DEMONSTRABLE / ACTIONABLE METRIC:** Attributed campaigns; impersonated officials; audience reach; correction latency; administrative or behavioral effects.

**PREVENTIVE CONTROL:** Minimize exploitable data; authenticate official communications; prebunk known narratives; train personnel; establish provenance and rapid coordination.

**FORENSIC / ASSURANCE CONTROL:** Archive content and official responses; attribute with confidence; measure reach, persistence, correction latency, behavioral effects, and administrative disruption.

**INDICATORS:** Attributed operations; reach; correction latency; impersonation; behavioral or administrative effect.

**PRINCIPAL GAPS:** G-01,02,03,06,07,09,10

**CONFIDENCE:** Existence HIGH / Recurrence MOD-HIGH

**TV-14 Illicit campaign finance and covert political support**

**EVIDENCE-BACKED BASELINE:** Covert funding, prohibited foreign support, straw donors, shell entities, and undisclosed relationships are documented election-crime pathways, but this assessment's current corpus is incomplete.

**WARNING JUDGMENT:** Financial access can fund influence, recruit domestic intermediaries, acquire data, or sustain operational infrastructure while obscuring foreign or prohibited sponsorship.

**DEMONSTRABLE / ACTIONABLE METRIC:** Prohibited contributions, straw-donor cases, beneficial-ownership gaps, referrals, disposition and funds traced.

**PREVENTIVE CONTROL:** Validate donor identity and payment source; identify beneficial ownership; screen foreign indicators; strengthen reporting and conflict controls.

**FORENSIC / ASSURANCE CONTROL:** Trace funds and entities; link referrals to dispositions; preserve campaign and payment records; deconflict repeated allegations from common origins.

**INDICATORS:** Funds traced; entity resolution; foreign indicators; referral and disposition; disclosure lag.

**PRINCIPAL GAPS:** G-01,02,03,05,06,07,08,10

**CONFIDENCE:** Existence MOD / Scale-effect LOW

**TV-15 Systemic, accidental, and cascading failure**

**EVIDENCE-BACKED BASELINE:** Human error, configuration mistakes, equipment failure, weather, power, communications, and staffing shortages regularly affect election operations without malicious intent.

**WARNING JUDGMENT:** Accidental failures can become cascading events or be exploited through cyber activity, obstruction, or false narratives; concentration can amplify local failure.

**DEMONSTRABLE / ACTIONABLE METRIC:** Outages, configuration errors, delayed openings, correction time, manual-contingency coverage, common-vendor concentration.

**PREVENTIVE CONTROL:** Testing, change control, redundancy, training, spare capacity, vendor diversification, and manual fallback reduce both accident and adversary opportunity.

**FORENSIC / ASSURANCE CONTROL:** Capture incidents, corrections, downtime, affected transactions, root cause, and independent outcome verification; publish lessons and corrective action.

**INDICATORS:** Outages; configuration errors; manual-contingency coverage; recovery time; concentration and recurrence.

**PRINCIPAL GAPS:** G-01,02,04,06,08,09

**CONFIDENCE:** Existence HIGH / Event-specific LOW-MOD

## 4. Integrated Assessment

### 4.1 Multiple real vectors coexist with an unresolved national prevalence picture

#### ASSESSMENT

**Multiple U.S. election-security threat vectors are demonstrably real, and several have been successfully exploited in documented cases. The available evidence does not support a reliable national ranking by frequency, undetected prevalence, coordination, or electoral effect.**

**Confidence: High.** Existence is supported through independent evidence classes: official national administrative data, judicial and prosecutorial records, direct investigative observation, technical guidance, official threat assessments, and controlled case repositories. National ranking is constrained by incomplete coverage, inconsistent definitions, purposive case selection, unknown detection probability, incompatible denominators, and uneven final-disposition reporting.

**Implication:** Decision-makers should not wait for national-level fidelity before addressing a demonstrated pathway. They should also resist using any single or bounded case as evidence that the pathway is widespread. The appropriate response is risk-proportionate prevention, measurement, and warning. Priority should be given to measures that lie entirely within the authority's ability to control, and which will have the maximum downstream benefit.

### 4.2 Election integrity begins before the first active input

#### ASSESSMENT

**The largest preventable upstream exposure is activation of registrations, users, software, devices, or transactions without sufficient evidence that each legally material condition has been satisfied.**

For voter registration, the standard should be affirmative eligibility assurance before ordinary activation whenever legally and operationally possible. The control should evaluate identity, citizenship where required, age, residence or domicile, jurisdiction, applicable disqualification, restoration, duplicate-active-record status, completeness, and attestation. It should preserve the original application, evidence, source, processing history, and decision.

**The same standard must protect eligible citizens.** An unresolved database mismatch is not evidence of ineligibility. A fair system uses more than one lawful evidence path, distinguishes missing information from contradiction, requires secondary review for consequential adverse action, provides specific notice and cure, preserves provisional participation where required, protects sensitive addresses and records, and supplies timely appeal. Security and access are coequal integrity requirements.

**Confidence: Moderate to high.** The control logic is strong; implementation authority, source availability, legal standards, data quality, and deadlines vary across jurisdictions. The recommendation therefore concerns a common assurance outcome and auditable process, not a single inflexible national database rule.

### 4.3 Paper availability is not the same as outcome assurance

#### ASSESSMENT

**The largest controllable technical gap is the difference between possessing paper evidence and using it to verify a material outcome independently of the original tallying system.**

More than 98 percent of jurisdictions report a paper ballot or auditable paper record [EAC-01; EAC-05]. That substantially improves the ability to reconstruct an election. It does not establish that the record is voter-verified, securely preserved, reconciled to ballot accounting, selected for a statistically meaningful audit, or capable of correcting an incorrect outcome before certification.

The National Academies recommends human-readable paper ballots and risk-limiting audits because they can provide evidence independent of software [NAS-01]. The EAC has developed voluntary audit resources and draft national standards, but audit methods and legal requirements remain jurisdiction-dependent [EAC-02; EAC-03].

**Confidence: High.** The distinction is supported by official administrative data and authoritative technical analysis.

## 4.4 Technology creates efficiency and common-mode risk

### ASSESSMENT

**Technology becomes most consequential when the same platform performs a critical function, records the event, and supplies the only evidence that it acted correctly.**

Registration systems, electronic pollbooks, election-management systems, scanners, ballot-marking devices, reporting systems, cloud services, managed providers, and campaign platforms improve speed, accessibility, and administrative capacity. They also create credential, vendor, software, logging, remote-access, supply-chain, and concentration risk. Electronic pollbooks were used by 39.9 percent of jurisdictions in 2024, while testing and certification practices were not uniform [EAC-01].

The appropriate policy objective is not categorical abandonment of technology. It is limiting sole-source dependence: independent paper or analog evidence, controlled records, authenticated logs, manual contingencies, and external verification must survive failure or compromise of the platform.

**Confidence: High** regarding vulnerability and assurance logic; **low** regarding successful alteration of certified U.S. vote totals through the cited technical pathways.

## 4.5 Foreign actors can exploit domestic weaknesses without changing votes

### ASSESSMENT

**The most plausible foreign exploitation may combine voter and political data, domestic vulnerabilities, authentic disputes, stolen information, deceptive amplification, and disruption rather than direct alteration of tabulated votes.**

CISA, FBI, and ODNI identify the PRC, Russia, and Iran as principal nation-state foreign malign-influence actors and describe tactics including false personas, proxies, hack-and-leak activity, synthetic content, and exploitation of domestic narratives [FMI-01; FMI-02]. The prior Tornberg PRC baseline assesses with high confidence that PRC intelligence treated U.S. voter, campaign, political, communications, and personal information as sustained collection priorities and possessed pipelines for targeting and influence support. It does not establish PRC alteration of ballots, tabulation, certification, or the 2020 result [TB-PRC-01].

**Confidence: Moderate to high** for continuing attempts to exploit information and administrative weaknesses; **low** for direct foreign alteration of certified vote totals in the reviewed record.

## 4.6 Detection capacity is itself a security control

### ASSESSMENT

**A system that cannot reliably show who accessed it, what changed, what moved through it, which controls activated, and how discrepancies were resolved cannot support strong negative judgments or rapid correction.**

Absence of reported incidents has different meaning when a pathway is highly observable than when activity can remain silent unless the legitimate voter participates, a cross-jurisdiction match occurs, a log is retained, or an official refers the matter. Standardized logging, reconciliation, evidence retention, referral tracking, disposition reporting, and control-performance measures would improve security and analytic confidence simultaneously.

**Confidence: High.** The principal uncertainty is the distribution and performance of current detection controls, not the value of observability itself.

## 4.7 Baseline effect

### REINFORCES

Election fraud, interference, administrative failure, cyber targeting, and influence activity are real classes of risk. Several pathways have been exploited.

### REFINES

The principal national problem is uneven assurance and limited measurement across a decentralized system—not evidence that all elections or certified counts are broadly compromised.

### CONTRADICTS

### PRESERVES

Claims that anomaly totals, duplicate registrations, allegations, referrals, charges, and convictions are interchangeable or can be used as one national fraud rate.

The public record reviewed does not establish nationwide compromise of certified vote counts. Warning and mitigation remain justified without making that claim.

## 5. Preventive and Forensic Assurance Architecture

The control objective is a layered system in which invalid or unauthorized inputs are prevented where possible, bypasses are detected and contained, material outcomes are independently verified, and failures produce actionable learning.

### 5.1 Eligibility assurance before ordinary registration activation

| CRITERION                             | MINIMUM CONTROL EXPECTATION                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity</b>                       | Resolve the applicant to a unique person using reliable, lawful evidence; distinguish exact identity, probable match, and unresolved conflict.                                          |
| <b>Citizenship</b>                    | For federal elections, support U.S. citizenship through legally authorized evidence. A stale immigration-status record should not override current citizenship evidence without review. |
| <b>Age</b>                            | Confirm eligibility under the jurisdiction's controlling date rule, including lawful preregistration where applicable.                                                                  |
| <b>Residence / domicile</b>           | Apply the state's legal standard; support nonstandard addresses, homelessness, military/overseas status, and protected addresses through defined alternatives.                          |
| <b>Jurisdiction</b>                   | Assign the correct state, locality, district, and precinct; identify conflicts before ballot issuance.                                                                                  |
| <b>Disqualification / restoration</b> | Use current controlling records and a defined escalation process when court, corrections, or state data disagree.                                                                       |
| <b>Duplicate active record</b>        | Search intrastate and available interstate resources before creating another active record; do not treat a probable match as an automatic merger or rejection.                          |
| <b>Application integrity</b>          | Confirm legally material fields, required attestation, signature, source, and worker/system provenance.                                                                                 |
| <b>Controlled disposition</b>         | Activate, hold pending, provide a conditional/provisional path, reject with notice, or refer suspected intentional falsification under defined authority.                               |

Legal note: the National Voter Registration Act, HAVA, state law, judicial rulings, and other applicable authorities govern registration, list maintenance, provisional voting, removal, notice, and timing. The framework requires jurisdiction-specific legal implementation [DOJ-03; EAC-06].

### 5.2 Fairness and special-case safeguards

| SAFEGUARD                        | CONTROL PURPOSE                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No single-database veto</b>   | A material adverse decision should not rest solely on a source known to be stale, incomplete, probabilistic, or unable to answer the legal question. |
| <b>Specific notice and cure</b>  | Tell the applicant which criterion is unresolved, the evidence at issue, acceptable alternatives, deadline, assistance, and available review.        |
| <b>Equivalent evidence paths</b> | Publish lawful alternatives rather than relying on ad hoc discretion. Apply the same substantive standard through different evidentiary routes.      |

|                                        |                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Secondary human review</b>          | Require documented review before consequential denial based on automated matching, conflicting records, or unclear legal status.     |
| <b>Provisional participation</b>       | Preserve timely participation without counting a ballot until eligibility is resolved where federal or state law provides that path. |
| <b>Appeal and correction</b>           | Provide review fast enough to matter for the election and retain the original and corrected record.                                  |
| <b>Privacy and protected addresses</b> | Validate eligibility without unnecessarily disclosing residence or protected personal data.                                          |
| <b>Accessibility and language</b>      | Provide accessible and legally required language-support channels without changing the substantive eligibility standard.             |
| <b>Continuity</b>                      | Accept timely applications during outages and preserve submission date while verification proceeds through a controlled fallback.    |

### 5.3 Dual-error standard

|                                                                                                                                                                         |                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FALSE ACCEPTANCE</b><br>An ineligible applicant, unauthorized user, invalid ballot, unapproved software component, or improper transaction enters the active system. | <b>FALSE REJECTION</b><br>An eligible citizen, authorized user, valid ballot, approved component, or lawful transaction is wrongly denied, delayed, or excluded. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

A credible control system measures and reduces **both error classes**. Optimizing only against false acceptance can wrongfully exclude eligible voters. Optimizing only against false rejection can permit invalid inputs. The objective is accurate, timely, auditable determination with bounded exceptions and review.

### 5.4 Lifecycle controls

| STAGE                               | PREVENT                                                            | DETECT                                                     | CONTAIN / CORRECT                         | INDEPENDENTLY VERIFY                                       |
|-------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------|
| <b>Registration and eligibility</b> | Criterion validation; provenance; deduplication; fair exceptions   | Conflict alerts; continuous maintenance; access monitoring | Cure, appeal, record resolution; referral | Application-to-ballot trace; voting-history reconciliation |
| <b>Ballot issuance and access</b>   | Eligibility revalidation; issuance suppression; identity control   | Duplicate request/issuance alert; tracking                 | Cancel/reissue; provisional handling      | Issue-return-accept-count reconciliation                   |
| <b>Voting and custody</b>           | Authorized access; sealed inventory; two-person movement           | Seal, route, access, and count exceptions                  | Segregate, preserve, investigate          | Paper record, chain of custody, ballot accounting          |
| <b>Technology and vendors</b>       | Certification; provenance; MFA; segmentation; remote-access limits | Logs, hashes, monitoring, anomaly detection                | Isolate, revoke, restore known-good state | Independent paper/log evidence; external audit             |
| <b>Tabulation and certification</b> | Logic/accuracy testing; role separation; defined authority         | Reconciliation; audit escalation; variance review          | Correct, recount, delay where lawful      | Outcome audit and preserved evidence                       |
| <b>Information environment</b>      | Authenticated channels; prebunking; protected data                 | Impersonation and narrative monitoring                     | Rapid verified correction; coordination   | Archived official record and measured response             |

### 5.5 Minimum evidence independent of technology

- **Human-readable ballot evidence.** The voter can verify the record of selections, and the record can be recounted without relying on the original software.
- **Ballot accounting.** Every issued, returned, accepted, rejected, spoiled, provisional, duplicated, and counted ballot is reconciled.
- **Physical custody.** Transfers, seals, storage, access, and exceptions are recorded and reviewable.
- **Independent audit.** The evidence examined and selection method do not depend on the same software that produced the tally.
- **Authenticated operational record.** Unique users, changes, software versions, configurations, and incidents are independently retained.

- **Manual continuity.** Critical functions can proceed or fail safely during platform, power, communications, or vendor failure.

## 6. Decision Considerations and Recommendations

### DECISION PRINCIPLE

Standardize assurance outcomes and evidence requirements without assuming that one centralized operating model is suitable for every jurisdiction. Use existing lawful federal, state, local, funding, certification, procurement, and voluntary-standard authorities as appropriate; identify where legislation or additional authority is required.

### 6.1 Decision considerations

| CONSIDERATION                   | MATERIAL TRADEOFF                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security and access             | Eligibility validation must prevent improper activation without converting missing or stale data into automatic disenfranchisement.                                                                                  |
| Federalism and authority        | Federal elections are administered through state and local systems. National assurance objectives may require state adoption, funding conditions, voluntary standards, legislation, or intergovernmental agreements. |
| Speed and accuracy              | Rapid reporting improves confidence but must not outrun reconciliation, audit, and certification evidence.                                                                                                           |
| Technology and resilience       | Technology can improve performance. The requirement is independent evidence and manual fallback—not reflexive removal of useful systems.                                                                             |
| Privacy and observability       | Logging and identity resolution improve security but increase privacy and data-protection obligations.                                                                                                               |
| Uniformity and local conditions | Common outcome standards should preserve lawful adaptations for rural, tribal, military, overseas, disability, language, protected-address, and emergency conditions.                                                |
| Cost and capacity               | Small jurisdictions may need shared services, grants, templates, and regional support to meet common assurance outcomes.                                                                                             |
| Transparency and security       | Public evidence improves trust, but technical details, personal data, and security-sensitive information require controlled disclosure.                                                                              |

### 6.2 Recommended actions

| ID   | RECOMMENDED ACTION                                                   | PURPOSE                                                                                                                                                                                                        | VECTORS        | ACTIONABLE METRICS                                                                                   | HORIZON                                 |
|------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------|-----------------------------------------|
| R-01 | Establish an outcomes-based national election-assurance framework    | Define minimum evidence, control, metric, incident, audit, and continuity outcomes across the election lifecycle. Permit lawful state implementation paths rather than prescribing one operating system.       | All vectors    | Framework adoption; jurisdiction coverage; unresolved exceptions                                     | 0–18 months                             |
| R-02 | Require pre-entry eligibility assurance with fair exception handling | Verify every legally material criterion before ordinary activation where lawful and possible; provide alternative evidence, secondary review, cure, appeal, privacy protection, and provisional participation. | TV-01–04       | Validation coverage; false acceptance/rejection; cure, appeal, reversal; resolution time             | Immediate design; phased implementation |
| R-03 | Make every material outcome independently reconstructable            | Require voter-verifiable paper evidence, complete ballot accounting, controlled custody, and preserved records sufficient for recount or audit independent of the original system.                             | TV-05,09–12,15 | Paper-evidence coverage; accounting completion; custody exceptions; reconstructability test          | Immediate / before certification        |
| R-04 | Adopt a common post-election outcome-audit standard                  | Use risk-limiting audits or a comparably defensible method where feasible; define escalation, correction, evidence, timing, and publication requirements before certification.                                 | TV-09–12,15    | Audit coverage; risk limit or equivalent; escalation and correction; completion before certification | 1–2 cycles                              |

| ID   | RECOMMENDED ACTION                                                         | PURPOSE                                                                                                                                                                                                      | VECTORS              | ACTIONABLE METRICS                                                                            | HORIZON                                    |
|------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------|
| R-05 | Set minimum election-technology and vendor assurance controls              | Unique credentials, MFA, least privilege, two-person control, segmentation, software provenance, hash validation, restricted remote access, independent logging, tested backups, and vendor incident duties. | TV-08-10,12,15       | Control coverage; exceptions; hash/config variance; remote sessions; recovery time            | Immediate for high-risk gaps; 12-24 months |
| R-06 | Standardize incident, referral, and control-performance reporting          | Create a common taxonomy separating allegation, anomaly, attempt, control activation, verified exploitation, acceptance, count effect, disposition, and correction.                                          | All vectors          | Reporting completeness; disposition closure; detection/correction latency; evidence retention | 0-18 months                                |
| R-07 | Reduce voter-data exposure and monitor authorized use                      | Minimize public and internal fields to lawful need; protect sensitive data; authenticate users; log access; contractually limit vendors and data brokers; investigate anomalous use.                         | TV-01,03,06,07,09,13 | Fields exposed; users/vendors; anomalous access; data-sharing compliance                      | Immediate and continuing                   |
| R-08 | Strengthen trusted-access and manual-continuity assurance                  | Inventory privileged roles; require dual control for high-consequence actions; train and supervise temporary workers; exercise manual fallbacks and cross-jurisdiction support.                              | TV-08,11,12,15       | Privileged-account coverage; dual-control exceptions; exercise success; staffing threshold    | Before each election                       |
| R-09 | Institutionalize foreign-influence warning and authenticated communication | Protect official channels, prebunk recurring narratives, monitor impersonation and hack-and-leak activity, coordinate attribution, and publish verified control outcomes quickly.                            | TV-09,13,14          | Authenticated-channel coverage; correction latency; reach; official coordination              | Immediate and continuing                   |
| R-10 | Create a controlled national evidence and disposition repository           | Maintain privacy-protected case, incident, control, audit, and final-disposition records to support trend analysis without converting allegations into findings.                                             | All vectors          | Jurisdiction coverage; deduplication; final-disposition rate; source independence             | 12-36 months                               |

Recommendations are traceable to the vector and gap register. They inform executive judgment and require legal, fiscal, operational, privacy, accessibility, and federalism review before adoption.

## 6.3 Immediate 2026 priorities

- **Close known pre-entry gaps.** Identify jurisdictions or systems activating registrations without documented evidence for each required criterion; implement fair secondary review and deadline-preserving fallback.
- **Verify independent evidence.** Confirm voter-verifiable paper, ballot-accounting, custody, audit, and recount readiness before voting begins.
- **Reduce privileged and vendor exposure.** Eliminate shared accounts, require MFA and least privilege, inventory remote access, validate software, and test log retention.
- **Exercise failure modes.** Run manual-continuity, cyber, reporting, custody, and public-communication exercises using realistic election deadlines.
- **Publish indicator owners.** Assign who receives, validates, escalates, and communicates each warning indicator.
- **Separate incident from effect.** Use one reporting template that identifies the evidence stage and prevents capability, anomaly, or allegation from being presented as a counted-ballot effect.

## 6.4 Measures of success

| OUTCOME              | MEASURE OF SUCCESS                                                                                                                               |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Input assurance      | Every active record and privileged user has documented support for required criteria and authority, or a controlled exception with review.       |
| Outcome assurance    | Every material contest can be reconstructed and verified from independent evidence before certification.                                         |
| Control transparency | Coverage, exceptions, false acceptance/rejection, audit, cure, and correction performance are reported in comparable form.                       |
| Detection            | Material access, custody, transaction, reporting, and influence events generate observable records with assigned review.                         |
| Resilience           | Critical functions continue safely or fail predictably during platform, vendor, power, communications, or personnel disruption.                  |
| Public confidence    | Officials can explain what happened, what evidence exists, what controls worked or failed, and what was corrected without overstating certainty. |

## 7. Indicators, Information Gaps, and Revision Triggers

### 7.1 Priority warning indicators

| DOMAIN                     | INDICATOR                                                                                                                                                           | PRIMARY OWNER                                              | CADENCE                              |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------|
| Registration / eligibility | Increase in activations lacking criterion evidence; unresolved duplicate-active records; high cure/reversal; database conflict clustered by source or jurisdiction. | State election authority; registration system owner        | Weekly in-cycle; daily near deadline |
| Ballot access / mail       | Unusual request, issue, return, rejection, cure, or address concentration; duplicate-ballot control activation; delivery anomaly.                                   | Local/state election authority; postal and system partners | Daily after issuance                 |
| Trusted access             | Shared credentials; unusual privilege change; disabled logs; after-hours or out-of-role actions; repeated dual-control exception.                                   | Election CIO/CISO; jurisdiction lead                       | Continuous / daily review            |
| Cyber / vendor             | Unauthorized access, hash mismatch, remote session, ransomware, multi-client outage, unexpected software or configuration change.                                   | CISA partners; state/local incident command; vendor        | Continuous; immediate escalation     |
| Custody / operations       | Broken seal, inventory mismatch, route deviation, unlogged transfer, unattended storage, staffing below safe threshold.                                             | Local election authority; law enforcement as appropriate   | At each transfer / shift             |
| Reporting / certification  | Large unexplained unofficial-to-canvass variance; repeated feed changes; unresolved reconciliation; missed deadline; threat against certifying official.            | State/local canvass and legal authority                    | Continuous after polls close         |
| Information operations     | Official impersonation, synthetic evidence, coordinated narrative onset after incident, hack-and-leak, foreign-linked amplification.                                | FBI/CISA/ODNI partners; state communications               | Continuous; rapid verification       |
| Finance / covert support   | Foreign-source indicators, shell entities, payment clustering, unexplained data or consultant payments, disclosure evasion.                                         | FEC/DOJ/state authorities as applicable                    | Periodic; immediate on threshold     |

## 7.2 Priority information gaps

| GAP                         | CURRENT LIMITATION                                                                                                                                     | COLLECTION / CONTROL REQUIREMENT                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| G-01 Completeness           | No representative national incident and disposition universe.                                                                                          | Create reporting coverage statement; distinguish unreported, unresolved, cleared, charged, convicted, corrected.            |
| G-02 Detectability          | Unknown probability that successful activity would create and preserve observable evidence.                                                            | Map expected observables, log coverage, referral thresholds, retention, and review ownership for each vector.               |
| G-03 Evidence bridge        | Many sources stop before ballot acceptance, count, certification, or effect.                                                                           | Use case-level stage matrices from attempt through effect; do not infer omitted stages.                                     |
| G-04 Control performance    | Coverage is often known; false-positive, false-negative, bypass, and recovery performance are not.                                                     | Measure control activations, misses, cure, reversal, audit escalation, and residual risk.                                   |
| G-05 Normalization          | State laws, offense elements, procedures, and dispositions differ.                                                                                     | Build a jurisdictional legal and procedural crosswalk before aggregating cases.                                             |
| G-06 Exposure / denominator | Case counts frequently lack the susceptible population or transaction set.                                                                             | Use vector-specific denominators: applications, active records, mail ballots, privileged users, systems, or custody events. |
| G-07 Attribution / intent   | A demonstrated mechanism does not identify a current actor, tasking, coordination, or objective.                                                       | Separate capability, access, preparation, authorization, action, and effect; identify discriminating evidence.              |
| G-08 Coverage balance       | Current evidence overrepresents conventional fraud and registration relative to cyber, insider, vendor, finance, certification, and influence vectors. | Balance collection across operational families and include failed attempts and near misses.                                 |
| G-09 Warning design         | Indicators lack common thresholds, owners, cadence, and response.                                                                                      | Assign observable condition, baseline, threshold, owner, reporting path, and required action.                               |
| G-10 Source integrity       | Derivative repetition, advocacy datasets, opaque algorithms, and mixed dispositions can inflate apparent support.                                      | Require provenance, definition, deduplication, independent support, reproducible calculations, and bounded use.             |

## 7.3 Conditions requiring reassessment

- **Direct effect evidence.** Verified evidence that an actor altered accepted ballots, tabulation, canvass, certification, or an election outcome.
- **Control-performance evidence.** National or representative data showing materially higher or lower prevention, detection, correction, or audit performance than currently understood.
- **Foreign operational bridge.** Evidence connecting foreign collection, access, tasking, authorization, action, and election-process effect.
- **Systemic common-mode failure.** A vendor, software, credential, or infrastructure event affecting multiple jurisdictions or compromising independent evidence.
- **Eligibility error evidence.** Demonstrated improper activation or improper exclusion at scale, or a control that materially shifts either error class.
- **Legal change.** Federal or state law, judicial rulings, audit standards, certification requirements, or data-access authorities materially change the control environment.
- **New evidence set.** A representative, independently controlled national incident or disposition dataset becomes available.
- **Adversary adaptation.** New tactics bypass current indicators or exploit a mitigation adopted in response to this assessment.

## 7.4 Shelf life

The vector-existence judgments are expected to remain useful through the 2026 election cycle unless direct contradictory evidence emerges. Exposure, control-coverage, legal, threat, and warning judgments should be reviewed at least quarterly and immediately following a material incident, major legal change, authoritative assessment, or significant new dataset.

## 8. Assumptions, Alternatives, and Limitations

### 8.1 Material assumptions

| ASSUMPTION                | EFFECT ON ASSESSMENT                                                                                                                                                                                                |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational horizon       | The 2026 federal election cycle through certification is the principal decision horizon; structural controls may extend through 2028.                                                                               |
| Public-source sufficiency | Public and declassified evidence is sufficient to establish vector existence and major control gaps, but not all actor intent, access, or classified detectability judgments.                                       |
| Existing authorities      | Recommendations can be pursued through some combination of current federal, state, local, procurement, certification, grant, and voluntary-standard authorities; specific legal authority requires separate review. |
| Control behavior          | Jurisdictions generally intend to comply with law and administer accurate elections; variance primarily reflects law, resources, systems, procedure, and implementation rather than presumed misconduct.            |
| Measurement integrity     | Official administrative data are generally the strongest national baseline, but their definitions, completeness, and reporting capabilities remain variable.                                                        |
| Adversary adaptation      | Domestic and foreign actors will adapt to visible controls and may shift toward lower-observability or influence pathways.                                                                                          |

### 8.2 Credible alternatives

|                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DECENTRALIZATION IS PRINCIPALLY A RESILIENCE ADVANTAGE</b><br><br>Distributed administration can limit nationwide common-mode failure and permit local adaptation. This alternative refines rather than overturns the assessment: the concern is uneven assurance and poor comparability, not decentralization itself.      | <b>ASSESSMENT EFFECT</b><br><br>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence. |
| <b>DOCUMENTED CASES OVERSTATE OPERATIONAL THREAT</b><br><br>Prosecution and advocacy datasets are selection-biased toward discovered and provable conduct. This is accepted for prevalence estimates. It does not negate demonstrated exploitability.                                                                          | <b>ASSESSMENT EFFECT</b><br><br>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence. |
| <b>CURRENT CONTROLS ALREADY REDUCE RESIDUAL RISK TO A LOW LEVEL</b><br><br>Rejections, cures, audits, paper records, reconciliations, and prosecutions show controls often work. The national residual-risk judgment remains unresolved because control performance and detectability are not consistently measured.           | <b>ASSESSMENT EFFECT</b><br><br>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence. |
| <b>STRICTER ELIGIBILITY VALIDATION WOULD CAUSE MORE IMPROPER EXCLUSION THAN IT PREVENTS</b><br><br>This is plausible if implementation relies on stale or single-source data. <b>The recommended dual-error standard, alternative evidence, cure, appeal, and provisional path are designed to test and contain that risk.</b> | <b>ASSESSMENT EFFECT</b><br><br>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence. |

|                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>TECHNOLOGY IMPROVES RELIABILITY MORE THAN IT INCREASES RISK</b></p> <p>This may be true for many functions. The recommendation is not to remove useful technology; it is to ensure independent evidence, controlled access, auditable change, and manual continuity.</p>                                                | <p><b>ASSESSMENT EFFECT</b></p> <p>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence.</p> |
| <p><b>FOREIGN INFLUENCE IS PRIMARILY RHETORICAL AND NOT AN ELECTION-SECURITY ISSUE</b></p> <p>Influence may not alter ballots, but it can affect participation, officials, certification, resource allocation, and public confidence. Those are material election-security effects even when vote totals remain accurate.</p> | <p><b>ASSESSMENT EFFECT</b></p> <p>Retained as a credible alternative or moderating condition; does not replace the principal judgment without new evidence.</p> |

8.3 Limitations

- **No classified or privileged access.** The assessment cannot independently evaluate nonpublic collection, source sensitivity, detection architecture, or ongoing investigations.
- **Preliminary source packet.** The full process summary, source register, case-level validation record, and method audit have not yet been produced for this broader national assessment.
- **Current administrative baseline.** The 2024 EAVS is the most comprehensive available national administrative dataset, but it predates the 2026 election cycle.
- **No complete national incident census.** The Heritage export and other case collections are purposive and contain mixed dispositions, overlapping categories, and inconsistent source depth.
- **No independent legal opinion.** Legal characterizations are analytical orientation only. Implementation requires counsel and jurisdiction-specific authority review.
- **No cost model.** Recommendations are not yet costed, staffed, sequenced by jurisdiction, or tested against appropriations and acquisition constraints.
- **No final review.** The product has not completed source audit, legal review, technical review, red-team review, tradecraft review, editorial proof, approval, or release authorization.

8.4 Human accountability

The recommendations inform executive judgment. An accountable human professional must validate the evidence, confidence, legal feasibility, implementation design, and proportionality of any action.

## Appendix A. Controlled Definitions and Confidence

| TERM                             | CONTROLLED DEFINITION                                                                                                                                                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Election integrity</b>        | The lawful inclusion of eligible voters and valid ballots; exclusion or correction of invalid participation; accurate administration, counting, canvass, certification, and preservation of evidence; and protection against improper denial of eligible participation. |
| <b>Voting integrity</b>          | The integrity of eligibility, registration, ballot access, ballot casting, acceptance, counting, and reconciliation, including both improper inclusion and improper exclusion.                                                                                          |
| <b>Threat vector</b>             | A pathway through which an actor, error, failure, or condition can exploit a vulnerability to affect election confidentiality, integrity, availability, lawful participation, administration, certification, or confidence.                                             |
| <b>Vulnerability</b>             | A weakness or condition that can be exploited or can contribute to failure. Its existence is not evidence that exploitation occurred.                                                                                                                                   |
| <b>Demonstrated exploitation</b> | A supported event in which conduct crossed from theoretical capability into observable action against the pathway. It does not establish prevalence or effect beyond the supported stage.                                                                               |
| <b>Warning</b>                   | A judgment that an observable condition or credible pathway requires attention, preparation, collection, or action before uncertainty is fully resolved.                                                                                                                |
| <b>Control activation</b>        | Evidence that a preventive, detective, corrective, or assurance control identified, rejected, cured, reconciled, contained, or corrected a condition. It is not automatically evidence of successful fraud.                                                             |
| <b>Independent verification</b>  | Confirmation using evidence or a process not dependent on the same system, software, person, or record source that produced the original result.                                                                                                                        |
| <b>Residual risk</b>             | Risk remaining after applicable controls operate, accounting for coverage, performance, bypass, detectability, and consequence.                                                                                                                                         |
| <b>National prevalence</b>       | The frequency or distribution of a condition across a defined national population using a representative or otherwise defensible collection design and denominator.                                                                                                     |

### Confidence standard

| LEVEL                  | USE                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HIGH</b>            | Strong evidence from authoritative, direct, or independently corroborated sources; definitions and evidence stage are sufficiently clear; remaining gaps are unlikely to reverse the bounded judgment. |
| <b>MODERATE</b>        | Credible evidence supports the judgment, but source, coverage, bridge, detectability, legal, or control-performance gaps could materially refine it.                                                   |
| <b>LOW</b>             | Evidence is limited, dependent, opaque, poorly normalized, or missing material bridges; the judgment should be treated as provisional and collection-driven.                                           |
| <b>NOT ESTABLISHED</b> | The reviewed record does not support the proposition at the required evidence stage. This is not identical to proving the proposition false.                                                           |

## Evidence status vocabulary

| STATUS                     | MEANING                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SUPPORTED</b>           | The source directly supports the bounded proposition stated.                                                                                          |
| <b>SUPPORTED / BOUNDED</b> | The source supports part of the proposition; downstream intent, scale, coordination, or effect remains inferential.                                   |
| <b>DIRECTLY OBSERVED</b>   | A participant, recording, or contemporaneous artifact captures the conduct; legal disposition and broader context may remain unresolved.              |
| <b>OFFICIAL ALLEGATION</b> | A charging, complaint, warrant, or investigative record contains the allegation; analytic weight depends on the underlying evidence and disposition.  |
| <b>CONTEXT</b>             | The source informs capability, doctrine, controls, priors, or collection but does not establish the event under assessment.                           |
| <b>CORRECTED</b>           | Evidence supports a narrower formulation and contradicts or fails to support a broader claim.                                                         |
| <b>QUARANTINED</b>         | The lead remains relevant, but provenance, definition, independence, methodology, calculation, or bridge evidence is insufficient for definitive use. |

## Appendix B. Selected Source Register

This register identifies the principal sources supporting the draft. It is not the full source packet. Inclusion does not imply adoption of every claim in a source.

| ID / SOURCE                                                                                                                                                                       | SOURCE CLASS                                       | PRIMARY USE                                                                                                                      | PRINCIPAL LIMITATION                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>EAC-01</b> U.S. Election Assistance Commission, 2024 Election Administration and Voting Survey, 30 June 2025.<br><a href="#">Direct access</a>                                 | Primary official national administrative survey.   | National exposure, registration transactions, voting methods, workforce, technology, cure, rejection, and paper-record measures. | Certified state/local submissions; definitions and completeness vary by item and jurisdiction.                        |
| <b>EAC-02</b> U.S. Election Assistance Commission, Election Audits Across the United States.<br><a href="#">Direct access</a>                                                     | Official policy and administrative synthesis.      | Establishes variation in audit types and absence of one national auditing standard.                                              | Describes policy and practice; does not measure audit performance nationally.                                         |
| <b>EAC-03</b> U.S. Election Assistance Commission, Draft Voluntary National Election Audit Standards and 2026 hearing materials.<br><a href="#">Direct access</a>                 | Official draft standard / public process.          | Shows federal movement toward common voluntary audit principles.                                                                 | Draft and voluntary; not evidence of universal adoption or performance.                                               |
| <b>EAC-04</b> U.S. Election Assistance Commission, Voluntary Voting System Guidelines and migration resources.<br><a href="#">Direct access</a>                                   | Official technical standards framework.            | System certification generations, voluntary federal framework, and migration context.                                            | Voluntary unless adopted through state law; legacy systems remain in use.                                             |
| <b>EAC-05</b> U.S. Election Assistance Commission, paper-based voting system resources and 2024 EAVS technology findings.<br><a href="#">Direct access</a>                        | Official technical / administrative source.        | Paper ballot or auditable paper-record coverage.                                                                                 | Paper availability does not establish voter verification, custody, or meaningful audit.                               |
| <b>EAC-06</b> U.S. Election Assistance Commission, Voter Lists, Registration, Confidentiality, and Voter List Maintenance.<br><a href="#">Direct access</a>                       | Official administrative guidance.                  | Continuous list maintenance, privacy, and registration-process context.                                                          | Guidance does not replace state law or establish misconduct prevalence.                                               |
| <b>EAC-07</b> U.S. Election Assistance Commission, Best Practices for Election Technology.<br><a href="#">Direct access</a>                                                       | Official technical guidance.                       | Hash validation, audit logs, unique credentials, two-person control, custody, tamper evidence, continuity.                       | Best practices are not proof of implementation or performance.                                                        |
| <b>NIST-01</b> National Institute of Standards and Technology, VTS 200-1, Cybersecurity Framework Election Infrastructure Profile, January 2024.<br><a href="#">Direct access</a> | Official consensus technical profile.              | Cyber-risk management across voter registration, voting systems, supporting technology, and service providers.                   | Framework guidance; local implementation and maturity vary.                                                           |
| <b>NAS-01</b> National Academies of Sciences, Engineering, and Medicine, Securing the Vote: Protecting American Democracy, 2018.<br><a href="#">Direct access</a>                 | Authoritative consensus study.                     | Human-readable paper ballots, independent evidence, risk-limiting audits, and technology-resilience logic.                       | Predates later technology and law changes; remains foundational, not a current implementation census.                 |
| <b>FMI-01</b> CISA, FBI, and ODNI, Securing Election Infrastructure Against the Tactics of Foreign Malign Influence Operations, April 2024.<br><a href="#">Direct access</a>      | Joint official threat and mitigation guidance.     | Foreign actor tactics, information manipulation, impersonation, synthetic content, and election-infrastructure mitigation.       | Public warning product; underlying intelligence and full detection picture are not exposed.                           |
| <b>FMI-02</b> ODNI Foreign Malign Influence Center, Primer on Foreign Malign Influence, Volume 2, October 2024.<br><a href="#">Direct access</a>                                  | Official intelligence-community public assessment. | Distinguishes influence from interference and describes foreign tactics and domestic laundering.                                 | Public-level source characterization is limited; effect measurement remains difficult.                                |
| <b>DOJ-01</b> U.S. Department of Justice, Election Crimes Branch.<br><a href="#">Direct access</a>                                                                                | Official prosecutorial orientation.                | Documented federal election-crime categories including voting fraud and campaign-finance crime.                                  | Not a national incidence or conviction dataset.                                                                       |
| <b>DOJ-02</b> U.S. Department of Justice, Armstrong paid-registration case, 2026.<br><a href="#">Direct access</a>                                                                | Official charging and plea-agreement account.      | Corroborates paid registration and false-address conduct documented by undercover video.                                         | Bounded case; does not establish unlawful ballot casting, counting, national scale, or all broader recorded activity. |
| <b>DOJ-03</b> U.S. Department of Justice, National Voter Registration Act and federal voting-rights guidance.<br><a href="#">Direct access</a>                                    | Official legal guidance.                           | Registration opportunity, list maintenance, timing, voter protection, and enforcement context.                                   | Legal application is fact- and jurisdiction-specific; not a substitute for counsel.                                   |
| <b>MIT-01</b> MIT Election Data + Science Lab, Interstate Voter Registration Data Matching.<br><a href="#">Direct access</a>                                                      | Academic methodological synthesis.                 | False positives, false negatives, field selection, thresholds, and ordinary causes of multiple records.                          | Contextual method source; does not validate a specific commercial or advocacy match total.                            |

| ID / SOURCE                                                                                                                                                                                     | SOURCE CLASS                                          | PRIMARY USE                                                                                                       | PRINCIPAL LIMITATION                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NCSL-01</b> National Conference of State Legislatures, Access to and Use of Voter Registration Lists.<br><a href="#">Direct access</a>                                                       | State-policy synthesis.                               | Variation in voter-file access, fields, confidentiality, and permissible use.                                     | Secondary legal synthesis; current propositions should be checked against official state law.                                                  |
| <b>CENSUS-01</b> U.S. Census Bureau, Voting and Registration.<br><a href="#">Direct access</a>                                                                                                  | Primary federal statistical context.                  | Participation and registration denominators by demographic and geography.                                         | Survey estimates; not evidence of individual fraud or election interference.                                                                   |
| <b>OK-01</b> James O’Keefe, testimony before the Senate Committee on Homeland Security and Governmental Affairs, 15 July 2026; underlying public video series.<br><a href="#">Direct access</a> | Direct investigative observation and sworn testimony. | Paid-registration, false identity, petition, voter-data access, and worker-incentive pathways.                    | DOJ materially corroborates the Armstrong conduct; not every recorded actor or broader claim has a public final disposition.                   |
| <b>DB-01</b> Heritage Foundation Election Fraud Database export reviewed 18 July 2026: 1,619 rows, 1982–2025.<br><a href="#">Direct access</a>                                                  | Purposive case-discovery repository.                  | Examples of adjudicated or officially determined conduct and within-dataset method frequencies.                   | Not representative, exhaustive, or denominator-compatible; mixed dispositions, repeated rows, overlapping categories, and uneven source depth. |
| <b>TB-PRC-01</b> Tornberg Consulting, July 2026 Trump Election Interference Intel Release: Baseline Analysis Impacts, 17 July 2026.                                                             | Prior controlled public-source synthesis.             | PRC voter-data collection, processing, influence capability, and bounded warning; no established vote alteration. | Secondary synthesis and prior Tornberg product; not independent corroboration beyond its underlying sources.                                   |
| <b>TB-METHOD-01</b> Tornberg Consulting, Time-Gated Evidence Deconfliction and companion Public Source Packet, 17 July 2026.                                                                    | Method and source-control products.                   | Evidence-stage separation, source deconfliction, quarantine, detectability, and baseline-warning architecture.    | Method source; not substantive evidence of election misconduct.                                                                                |

Source register status: preliminary. Full source packet should record access dates, archived copies, file hashes where appropriate, claim-to-source crosswalks, independence, corrections, case dispositions, and quarantined claims.

## Appendix C. Review and Release Requirements

### UNQUALIFIED RELEASE IS NOT AUTHORIZED

This draft must not be represented as a final or approved Tornberg Consulting product, government assessment, legal opinion, or final client-facing intelligence product until such time as the following have been verified by an approving authority.

| REVIEW                     | MINIMUM REQUIREMENT                                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source audit               | Verify every material quantitative claim, URL, quotation, case disposition, source independence, definition, and limitation. Reconcile the EAVS figures used in tables with the official report.                   |
| Legal review               | Assess NVRA, HAVA, UOCAVA, Voting Rights Act, Privacy Act and state-law implications; identify which recommendations require legislation, rulemaking, funding conditions, procurement changes, or state adoption.  |
| Technical review           | Validate cybersecurity, voting-system, pollbook, audit, logging, vendor, and continuity statements with qualified election-technology professionals.                                                               |
| Tradecraft review          | Test evidence-stage separation, confidence, alternatives, detectability, source dependence, metric validity, and recommendation traceability.                                                                      |
| Red-team review            | Challenge the framing, selection of vectors, assumption that uneven assurance is the principal systemic issue, and the potential for recommendations to create improper exclusion or centralized common-mode risk. |
| Editorial and visual proof | Verify pagination, table breaks, headings, citations, brand use, accessibility, contrast, print behavior, and consistency across executive judgments and appendices.                                               |
| Approval authority         | An accountable senior professional must approve the final source packet, assessment, recommendations, handling, audience, and release version.                                                                     |

### Draft conclusion

The United States does not need proof of nationwide compromise before addressing demonstrated pathways. It needs disciplined separation of what is known from what is warned, strong preventive assurance before active inputs, independent evidence after transactions, measurable controls, fair procedures for exceptional cases, and a national ability to identify, contain, explain, and correct failure. The current record supports that direction with moderate-to-high confidence while leaving national prevalence, residual risk, and effect unresolved.

### FINAL STATUS

DRAFT FOR REVIEW, COORDINATION, AND REVISION.